Assignment 1 – Drone Technology Fundamentals & Components

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Q1. Drone Selection Challenge

A government agency wants to map a 500-acre agricultural field with long flight endurance, large area coverage, and accurate GPS navigation. For this requirement, I would recommend a **Hybrid VTOL (Vertical Take-Off and Landing) drone**.

Comparison of Drone Types

• Parameter	• Multirotor	• Fixed-Wing	• Hybrid VTOL
• Flight Time	• 15–30 min	• 60–120 min	• 45–90 min
• Coverage Area	• Low	• Very High	• High
• Hovering Capability	• Yes	• No	• Yes
• GPS Navigation	• Good	• Excellent	• Excellent
• Operational Cost	• Low	• Medium	• High
• Takeoff/Landing	• Vertical	• Runway Required	• Vertical
• Payload Capacity	• Medium	• Low-Medium	• Medium

Technical Analysis

1. Multirotor Drone

Multirotor drones are easy to operate and can hover accurately over a location. They are suitable for inspection and photography tasks. However, their limited battery life and smaller coverage area make them inefficient for mapping a large 500-acre farm.

2. Fixed-Wing Drone

Fixed-wing drones have excellent flight endurance and can cover very large areas in a single flight. They are highly efficient for mapping missions. However, they require a runway or launcher for takeoff and landing and cannot hover over a specific location.

3. Hybrid VTOL Drone

Hybrid VTOL drones combine the advantages of multirotor and fixed-wing aircraft. They can take off and land vertically like a multirotor while achieving long flight times and large-area coverage similar to a fixed-wing drone. They also provide excellent GPS-assisted autonomous navigation.

Recommended Drone: Hybrid VTOL

For mapping a 500-acre agricultural field, the Hybrid VTOL drone is the best option because:

- Long flight endurance reduces the number of battery changes.
- Large area coverage improves mission efficiency.
- Vertical takeoff and landing eliminate runway requirements.
- High GPS accuracy ensures precise field mapping.
- Suitable for autonomous survey missions in agricultural environments.

Q2. Component Compatibility Analysis

Given Drone Configuration

• **Component**	• **Specification**
• Frame	• 450 mm Quadcopter
• Motor	• 2212 – 1000KV BLDC
• Propeller	• 1045
• Battery	• 3S 2200mAh 30C LiPo
• ESC	• 30A

Why These Components Are Compatible

1. Frame and Motor Compatibility

The 450 mm frame is designed to support 2212-size brushless motors. The motor mounting holes and thrust requirements match the frame dimensions.

2. Motor and Propeller Compatibility

A 1000KV motor performs efficiently with a 1045 propeller on a 3S battery. This combination provides sufficient thrust while maintaining stable current consumption.

3. Battery and Motor Compatibility

A 3S LiPo battery supplies approximately 11.1V, which is suitable for a 1000KV motor. It delivers enough power without causing excessive motor speed.

4. ESC and Motor Compatibility

A 30A ESC can safely handle the current drawn by the 2212–1000KV motor with a 1045 propeller. This ensures reliable operation and prevents overheating.

5. Battery Discharge Capability

Battery Capacity = 2200mAh = 2.2Ah

Maximum Current = Capacity × C-Rating

= 2.2 × 30

= 66A

The battery can safely deliver up to 66A, which is sufficient for the four motors and ESC system.

• **Scenario**	• **Expected Problem**
• Smaller ESC (10A or 20A instead of 30A)	• ESC may overheat, fail, or burn because it cannot handle the motor current demand.
• Higher KV Motor (2300KV instead of 1000KV)	• Excessive RPM, higher current consumption, reduced flight time, overheating, and instability.
• Smaller Propeller (8045)	• Lower thrust generation, reduced lifting capability, unstable flight.
• Larger Propeller (1147)	• Excessive motor load, overheating, higher current draw, and possible ESC damage.

Commercial Drone Used in India

Drone: DJI Matrice 350 RTK

Application: Surveying, mapping, agriculture monitoring, infrastructure inspection.

Key Specifications:

- Flight time up to approximately 55 minutes
- RTK-based high-precision positioning
- Multiple payload support
- Advanced obstacle sensing
- Long communication range

Why Suitable:
The Matrice 350 RTK provides highly accurate GPS positioning, long endurance, and reliable operation. These features make it ideal for large-scale agricultural mapping and government survey projects.